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Robot path planning by integrating improved A* algorithm and DWA algorithm

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ABSTRACT: Traversing many nodes, low search efficiency, path slanting through obstacles, many path redundancy points, and the inability of dynamic real-time obstacle avoidance are all typical problems of A* algorithm. In this paper, A robot path planning method combining improved A* algorithm and DWA algorithm is proposed.The ratio of the distance between the current point and the starting point and the target point is introduced into the heuristic function, and the search efficiency is improved through the optimized heuristic function; the neighborhood search principle is simplified to 5-neighborhood search, and the nodes adjacent to obstacles are deleted to reduce the traversal nodes while avoiding diagonal penetration of obstacles; redundant node improvement method is used to reduce the degree of path transition; finally the path key points of the improved A* algorithm are extracted and DWA algorithm is fused to realize the robot to avoid unknown obstacles. Finally, experiments are carried out on Matlab to prove the feasibility of the new algorithm.

1. INTRODUCTION

Robotics-related applications have been developing rapidly, and the role of humans is being replaced by robots in certain settings and environments [1]. The prerequisite for a robot to be able to replace humans is to achieve autonomous navigation, which means that the robot can move autonomously from one point to another in an unstructured environment without human involvement and without collision. Autonomous robot navigation is divided into four modules, namely positioning, map building, path planning, and motion control, which complement each other to complete the robot's navigation work.

Path planning refers to the robot finding the best or second-best safe path from one point to another under limited conditions (such as the shortest path length, minimum energy consumption of the mobile robot and shortest time used by the algorithm) on the basis of known environmental information. Its core idea is to find the best or second best solution to a problem under limited conditions[2~3].The common methods for path planning are the Dijkstra algorithm, A* algorithm [4~6], RRT algorithm [7~8], ant colony algorithm [9~10], dynamic window method [11~12], etc. Many access nodes, low efficiency, many path turning points, and inability to avoid unknown obstacles are unavoidable problems of A* algorithm. To solve the problem of real-time obstacle avoidance, the local path planning algorithm is integrated into A* algorithm [13~18].

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In this paper, the heuristic function, search neighborhood and redundant nodes of A* algorithm are improved to speed up the search and decrease of the amount of turning points. Furthermore, the improved A* algorithm is fused with DWA algorithm to avoid unknown obstacles.

2. A* algorithm improvement

2.1 Traditional A algorithm*

Based on Dijkstra algorithm, A* algorithm is a heuristic search algorithm based on BFS algorithm. It is suitable for global path planning in static environment with known global map. It is assumed that S is the start node, T is the target node and C is the current node.It starts from S, uses the cost estimation function to estimate the cost of each node around C, selects the node with the least cost as the next search node, and continuously searches until T. Its cost estimate function is:

$$f(n)=g(n)+h(n) \tag{1}$$

where f(n) is the distance estimation from S to T, g(n) is the actual distance from S to C, and h(n) is the estimate of distance from C to T, called the heuristic function. h(n) can be used as Manhattan distance, Euclidean distance, and Chebyshev distance, etc. The h(n) is as follows

$$h(n)=\sqrt{(x_n-x_g)^2+(y_n-y_g)^2} \tag{2}$$

where x_n, x_g are the horizontal coordinates of C and T; y_n, y_g are the ordinates of C and T.

2.2 Heuristic function improvement

When the actual distance between C and T is larger than h(n), the number of nodes to search increases, the search area increases, and the efficiency decreases, but the path is shortest, when h(n) is equal to 0, it evolves into Dijkstra's algorithm. When the two are equal, the search is carried out by the shortest path, and the search efficiency is the highest. The actual distance between C and T is less than h(n), the number of nodes searched decreases, the search area decreases, and the search speeds up, but the results are not necessarily for the best. When h(n) is far more than the actual distance, it evolves into the BFS algorithm. Therefore, a reasonable choice of h(n) can ensure both finding the shortest path and high search efficiency.

In this paper, h(n) is improved so that it can be dynamically adjusted in different positions. The core of the improvement is to change the proportion of h(n) according to the position of C in the whole map. When C is close to S, a larger h(n) should be selected to reduce the search node and search scope. When C is close to T, to ensure the optimal path planning, add search nodes, and select a smaller h(n).

The improved h(n) is as follows

$$h(n)^*=\left(1+\frac{1}{L}\right)*h(n) \tag{3}$$

where l, L are the distance from C to T and S respectively.

2.3 Search Neighborhood Improvement

The traditional A* algorithm searches the surrounding eight neighborhoods of C, i.e., the top, bottom, left, right, and left of C as well as the eight nodes on the top, bottom, left, and right, as shown in Figure 1. The 8-neighborhood search makes more search nodes and reduces the search efficiency, so this paper changes the 8-neighborhood to 5-neighborhood for search. Assume that the connection between C and T is line m, and the vertical line of C is line n. The reserved node is selected according to the Angle difference between the two lines. Table 1 shows the 5-neighborhood search rules.

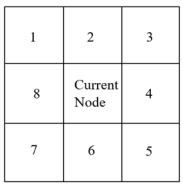


Figure 1. 8 Neighborhood map.

Table 1. 5-neighborhood search rules.

Clamping angle	Reserved Nodes	Abandoned Nodes
[0, 22.5°)∪[337.5°, 360°)	1, 2, 3, 4, 8	5, 6, 7
[22.5°, 67.5°)	1, 2, 3, 4, 5	6, 7, 8
[67.5°, 112.5°)	2, 3, 4, 5, 6	7, 8, 1
[112.5°, 157.5°)	3, 4, 5, 6, 7	8, 1, 2
[157.5°, 202.5°)	4, 5, 6, 7, 8	1, 2, 3
[202.5°, 247.5°)	5, 6, 7, 8, 1	2, 3, 4
[247.5°, 292.5°)	6, 7, 8, 1, 2	3, 4, 5
[292.5°, 337.5°)	7, 8, 1, 2, 3	4, 5, 6

The conventional A* algorithm does not consider the robot volume problem, and there is a phenomenon that the path slants through the obstacle vertices. In practical applications, the diagonal crossing of obstacles will lead to a collision and cause safety problems. Therefore, this paper optimizes the search neighborhood on the basis of 5 neighborhoods, so as to avoid diagonal penetration of obstacles. When the obstacle is in node 2, nodes 1 and 3 are deleted; when the obstacle is in node 4, nodes 3 and 5 are deleted; when the obstacle is in node 6, nodes 7 and 5 are deleted; when the obstacle is in node 8, nodes 1 and 7 are deleted.

2.4 Redundant node improvement

The path calculated by the traditional A* algorithm has a large number of redundant nodes, which leads to many path inflection points and unsmoothness, which is not conducive to robot movement.

There are two types of redundant nodes, including collinear nodes and redundant nodes. Firstly, the collinear nodes in the path are identified from the starting node and deleted, and then the redundant nodes are optimized to reduce the inflection points. From the starting node, the three adjacent nodes A, B, and C in the path, calculate the angle between the line AB and AC. If the angle is 0, the three nodes are co-linear, delete the co-linear node B, and so on, delete all the co-linear nodes in the path to get the new path A—>C—>E—>F—>I, as shown in Figure 2. Start again from the starting node, connect each inflection node, connect AE, AF, and AI, compare, and find that AE does not cross the obstacle and the path is less than A->C->E, so the path is updated to A-> E->F->I. Then start from node E, connect each inflection point, connect EI, the path E-> I is smaller than E-> F-> I and does not cross the obstacle to get the new path A-> E-> I.

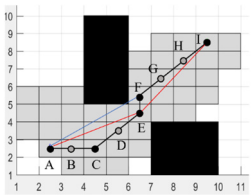


Figure 2. Redundant node deletion.

3. Dynamic window method

For unpredictable dynamic obstacles, the dynamic window method (DWA) can better solve this problem of dynamic obstacle avoidance. The DWA algorithm calculates the corresponding trajectory according to the robot's motion model, determines the velocity sampling space, samples multiple groups of linear

and angular velocities in the velocity space, predicts the trajectory of the next time interval according to the robot's kinematics model, and evaluates the predicted trajectory through the evaluation function, so as to obtain a safer and smoother optimal local path.

3.1 Robot motion model

The motion trajectory of the robot in a short period of time can be approximated as uniform linear motion, assuming that the robot can only move forward and rotate, and the forward velocity direction remains the same as the robot heading angle, then the robot motion model is

$$\begin{cases} x(t)=x(t-1)+v(t)*\Delta t*\cos\big(\theta(t-1)\big) \\ y(t)=y(t-1)+v(t)*\Delta t*\sin\big(\theta(t-1)\big) \\ \theta(t)=\theta(t-1)+\omega(t)*\Delta t \end{cases} \tag{4}$$

3.2 Robot velocity sampling

In the velocity (v, ω) sampling space, there are infinitely many sets of velocities, but the sampling velocities can be controlled within a certain range according to various constraints [19].

Affected by its own hardware, the actual speed range of the robot is expressed as

$$v_m=\{v\epsilon[v_{min},v_{max}],\omega\in[\omega_{min},\omega_{max}]\} \tag{5}$$

Affected by motor torque, the robot cannot reach the theoretical limit acceleration, so the actual velocity can be expressed as

$$v_d=\{(v,\omega)|v\in[v_c-\dot{v}_b*\Delta t,v_c+\dot{v}_a*\Delta t]\cap\omega\in[\omega_c-\dot{\omega}_b*\Delta t,\omega_c+\dot{\omega}_a*\Delta t]\} \tag{6}$$

For safety reasons, ensure that the robot stops before hitting the obstacle at maximum deceleration, and the velocity to meet the following conditions

$$v_a=\{(v,\omega)|v\le\sqrt{2dist(v,\omega)*\dot{v}_b}\cap\omega\le\sqrt{2dist(v,\omega)*\dot{\omega}_b}\} \tag{7}$$

3.3 Evaluation function

It consists of azimuth evaluation, obstacle distance evaluation, and current velocity evaluation. It is expressed as

$$G(v,\omega)=\sigma(\alpha*heading(v,\omega)+\beta*dist(v,\omega)+\gamma*velocity(v,\omega)) \tag{8}$$

where, heading(v, ω) is the azimuth evaluation function, which indicates the angular difference between the robot's trajectory end direction and the target point at the current set speed to ensure that the robot's orientation continuously tends to the end direction; dist(v, ω) is the obstacle distance evaluation function, which indicates the distance between the robot's predicted trajectory end and the nearest obstacle, to ensure the robot's obstacle avoidance ability; velocity(v, ω) is the current velocity evaluation function, which indicates the robot's current linear velocity. σ is the normalized smoothing coefficient, and α, β, γ are the weighting coefficients.

4. Fusion algorithm

The A* algorithm can achieve global path planning, but it has many search nodes, low search speed, many path turning points, unsmooth paths, and oblique obstacles. Although the improved A* algorithm has been improved to a large extent, it is still a static path-planning algorithm that cannot avoid obstacles when encountering dynamic obstacles or unknown static obstacles. Dynamic window method can avoid obstacles safely and effectively in unknown environment, and has the advantages of small computation, rapid response, and operability. Therefore, this paper proposes robot path planning by fusing the improved A* algorithm and DWA algorithm. Firstly, the key nodes in the global path are extracted, and then the local path is quickly updated based on this as the premise of the DWA algorithm. Finally, the robot can avoid obstacles in the real environment, as shown in Figure 3.

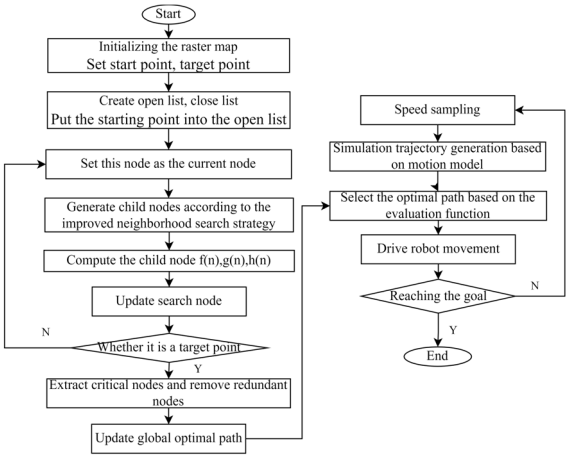


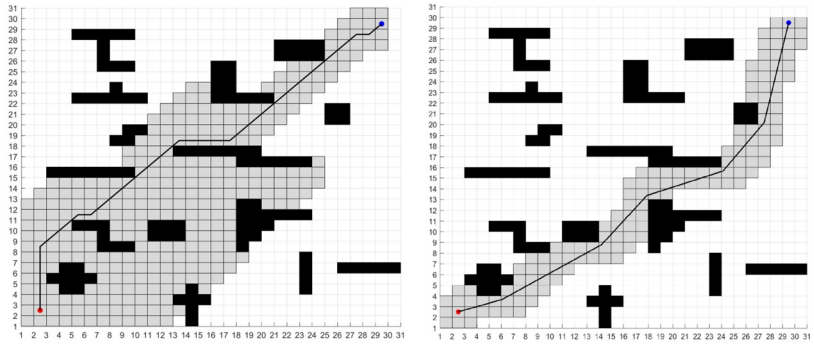
Figure 3. Flow chart of fusion algorithm.

5. Simulation experiments

5.1 Improved A algorithm simulation experiments*

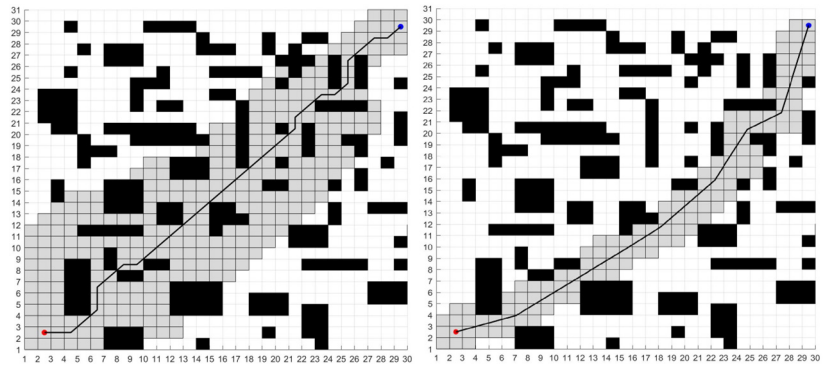
Build a simple raster map of size 30×30 and a complex raster map in Matlab. The black part of the map represents the obstacles, the gray part represents the access nodes, the solid red circle represents the starting node, and the solid blue circle represents T.

The planned path of the A* algorithm before and after improvement in the 30×30 simple raster map is shown in Figure 4, and the planned path in the 30×30 complex raster map is shown in Figure 5. Table 2 shows the performance parameters.



(a)Traditional A* algorithm (b) Improving the A* algorithm

Figure 4. Global path planning under a simple raster map.



(a)Traditional A* algorithm (b) Improving the A* algorithm

Figure 5. Global path planning under complex raster maps.

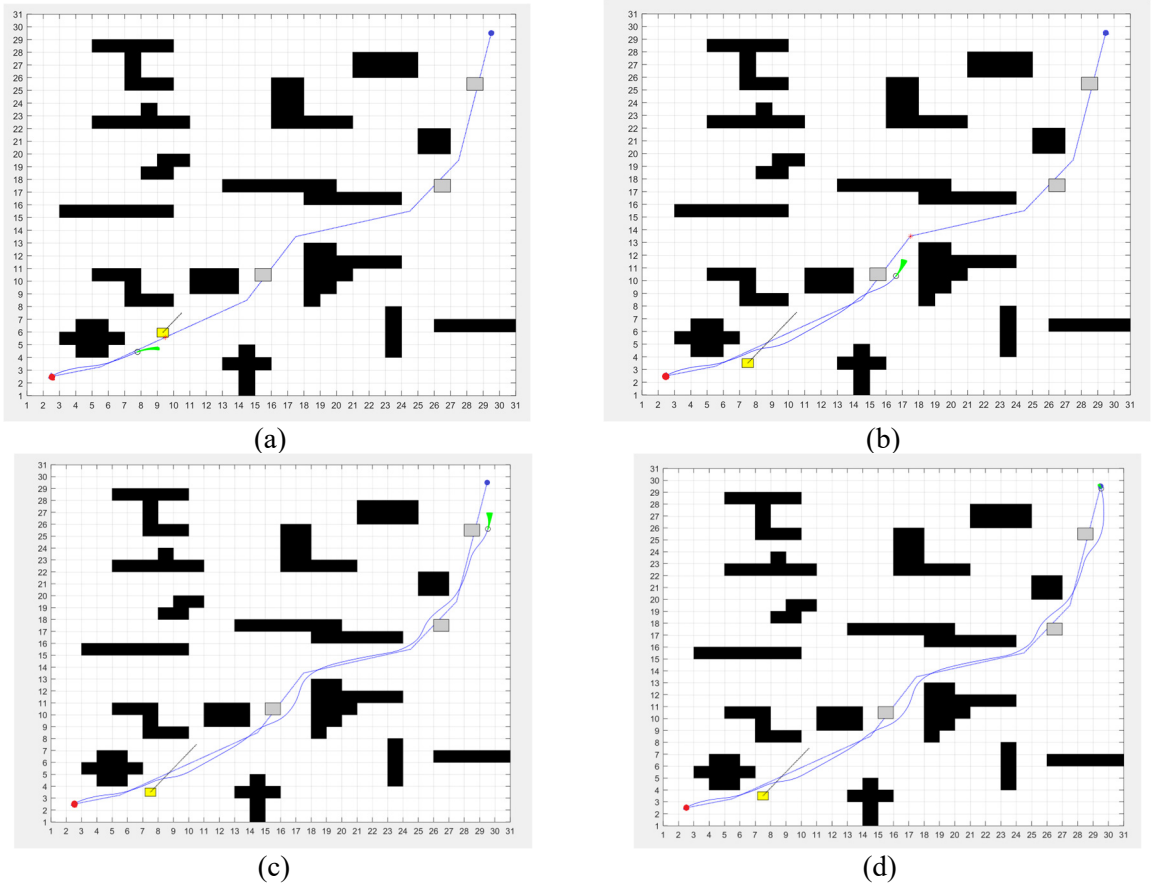
Table 2. Performance parameter benchmarking.

	Running time/s	Path length	Number of nodes traversed
Simple Raster Map Traditional A* algorithm	0.1656	41.6985	327
Simple Raster Map Improved A* algorithm	0.1283	41.0742	133
Complex raster maps Traditional A* algorithm	0.2530	41.1127	292
Complex raster maps Improved A* algorithm	0.2140	40.2834	125

By analyzing Table 2, in the simple environment, the improved A* algorithm reduces the running time by 22.52% and the number of traversed nodes by 59.33%; in the complex environment, the A* algorithm in the running time by 15.42% and the number of traversed nodes by 57.19%. Therefore, the improved algorithm speeds up the search. According to Figures 4 and 5, the path planned by the traditional A* algorithm slants through obstacles and cannot guarantee robot's safety. The improved A* algorithm has no significant reduction in path length but no slant through obstacles, which ensures the safety of the robot. Comparing the number of turning points of the paths planned by the improved A* algorithm before and after the improvement, the number of turning points decreases from 7 to 5 in the simple raster map; in the complex map, the turning point was reduced from 13 to 4.

5.2 Simulation experiment of the fusion algorithm

To verify whether the fusion algorithm can avoid obstacles in time, this paper adds 1 dynamic obstacle and 3 static obstacle to the 30×30 simple raster map. Yellow squares represent dynamic obstacles, gray squares represent static obstacles, red circles represent starting points, and blue circles represent target points, as shown in Figure 6.



(a) (b) (c) (d)

Figure 6. Simulation process of fusion algorithm.

According to the matlab simulation process diagram, the path planned by the fusion algorithm can effectively avoid dynamic obstacles and unknown static obstacles. And compared with the original planned path, the real path is smoother.

6. Conclusion

To solve the problems of A* algorithm with many visited nodes, low search efficiency, many path inflections points not smooth, path slanting through obstacles, and unable to avoid unknown obstacles, this paper proposes a path planning method for robots integrating improved A* algorithm and DWA algorithm. On the basis of the A* algorithm, the position proportion of the current node in the whole path is introduced into the heuristic function; the 8-neighborhood search is changed to a 5-neighborhood search, and the nodes adjacent to obstacles are deleted, which improves the search speed while avoiding the oblique crossing of obstacles; the redundant node improvement method is proposed, so as to ensure the smoothness of the path; the improved A* algorithm is fused with the DWA algorithm to achieve dynamic obstacle avoidance of the global path. Matlab simulation experiments show that the fusion algorithm proposed in this paper visits fewer nodes, has higher search efficiency, has smoother paths, and can effectively avoid obstacles in the environment.

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